86 400 result rows (test F1-macro of every run)
factors: data set, method, learner, rule, K; replicates: 5 folds
Q1. How much does each factor matter?
Q2. Do the levels differ when data sets are the units?
Q3. Is each method different from the baseline, by how much?
All 86 400 runs
Factorial ANOVA, type II sums of squares: 5 main effects, 6 interactions
Effect sizes ω^2 and partial η^2 per term
Residual checks; the same model on rank-transformed F1 (robustness)
Mean F1 per data set and level: a 9 × k table
Friedman test with the Iman-Davenport correction
Nemenyi post hoc test; critical difference (CD)
CD diagrams: methods, learners, rules, K
8 640 matched pairs per method (same data set, fold, K, learner and rule)
Wilcoxon signed-rank test, Holm correction over the nine methods
Effect size: matched-pairs rank-biserial r_{rb}
Mean difference with 95% cluster-bootstrap CI (data sets resampled); checked with BCa, t, Bonferroni
A difference counts as established only if it holds with data sets as the units (Nemenyi test or a bootstrap CI that excludes zero); ω^2, the CI and r_{rb} give its size
Answers to the research questions RQ1 to RQ4